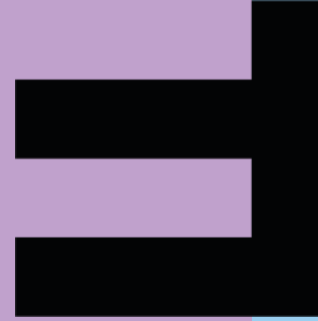




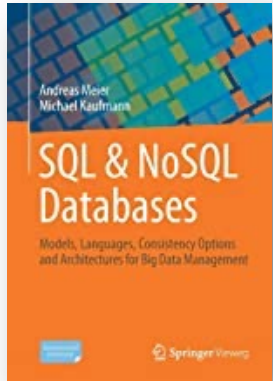
Technical Information Systems

6. Big Data and NoSQL-Systeme

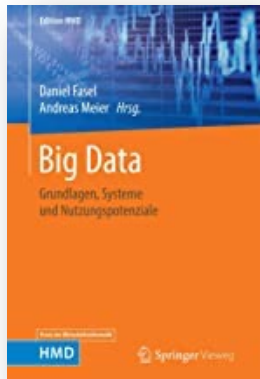
“Big data is at the foundation of all of the megatrends that are happening today, from social to mobile to the cloud to gaming.” — Chris Lynch

Literature

Additional literature on the topic



- [MeKa19]
MEIER, ANDREAS ; KAUFMANN, MICHAEL: *SQL & NoSQL databases*. New York, NY : Springer Berlin Heidelberg, 2019. URL <https://go.exlibris.link/J535GBxM>
— ISBN 978-3-658-24548-1



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Fasel, D. ; Meier, A. (Hrsg.): *Big Data: Grundlagen, Systeme und Nutzungspotenziale, Edition HMD* : Springer Vieweg, 2016.
URL <https://go.exlibris.link/9vtMd61l>
— ISBN 978-3-658-11588-3

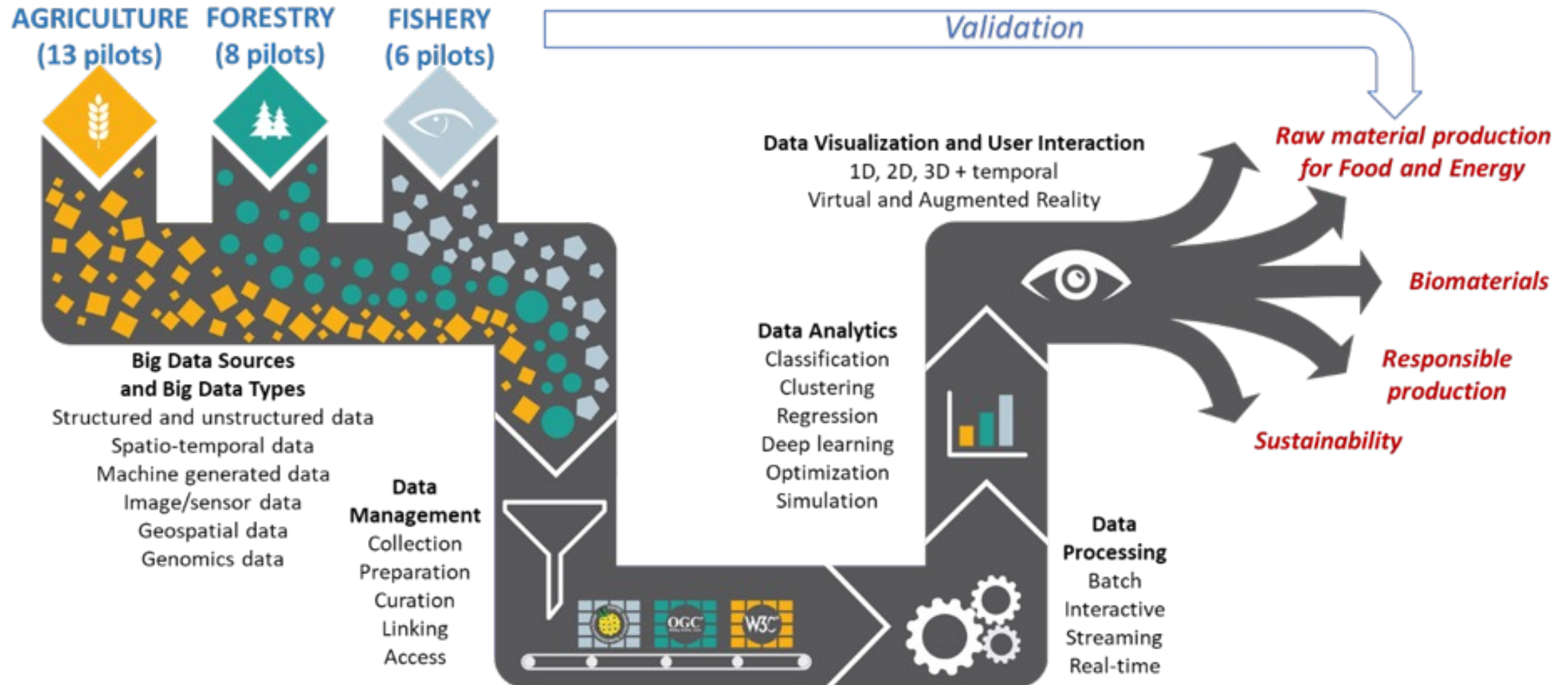


Big Data und NoSQL-Systeme

- Big Data

Example: The Databio Project

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<https://www.databio.eu/en/about-databio/summary/>

Exercise

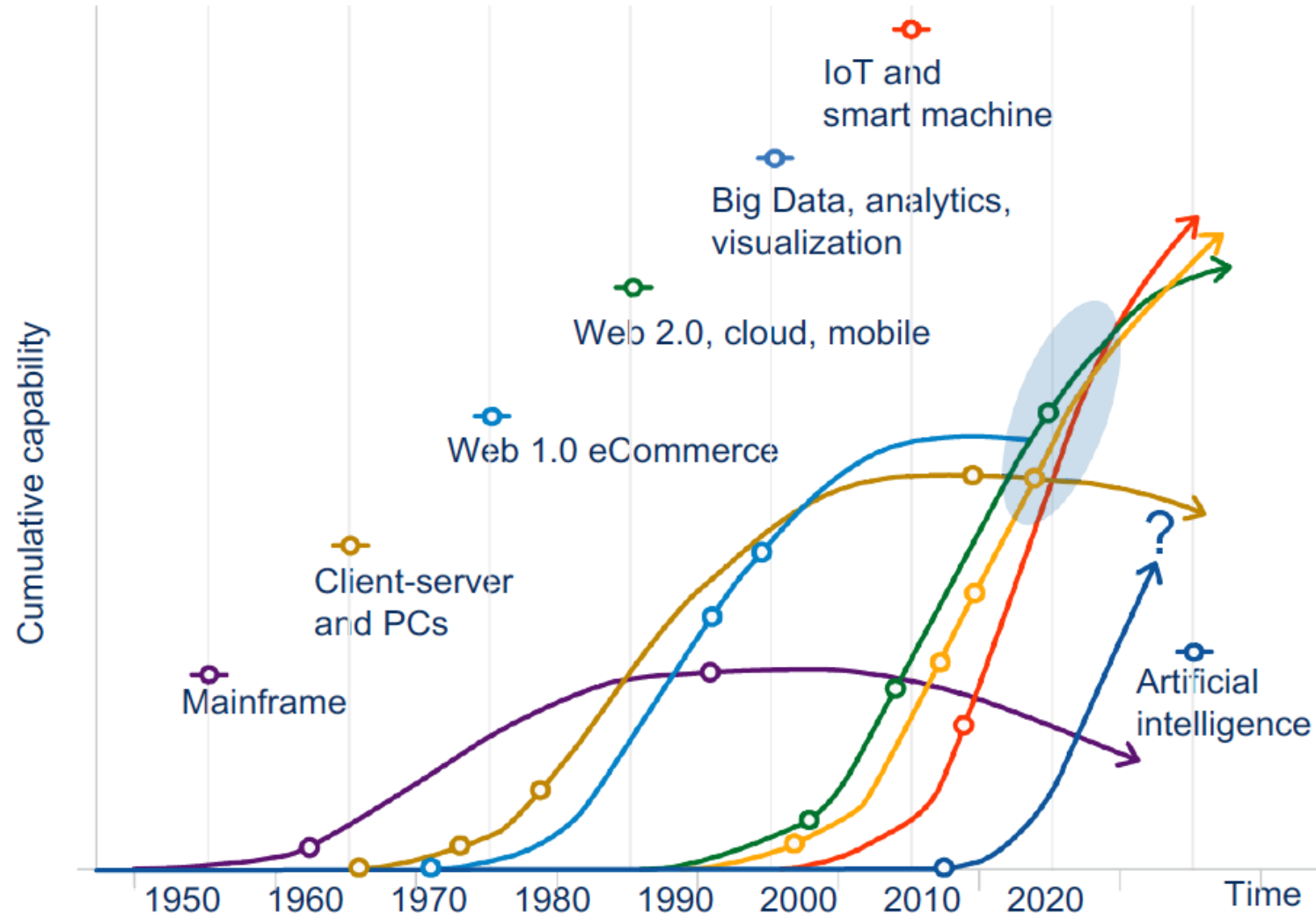
Computer Laws

- What is Moore's law? → **Computing power**
 - Computing power will double every 18 months (originally: number of transistors on an IC)
- What is Butter's law? → **Communication speed**
 - The amount of data communicated through a single optical fiber doubles every nine months.
- What is Kryder's law? → **Storage capacity**
 - the amount of data stored per centimeter square of a hard drive will double every 13 months (reduced to ~16 months)



The Combinatorial Effects of Technology are Accelerating Change

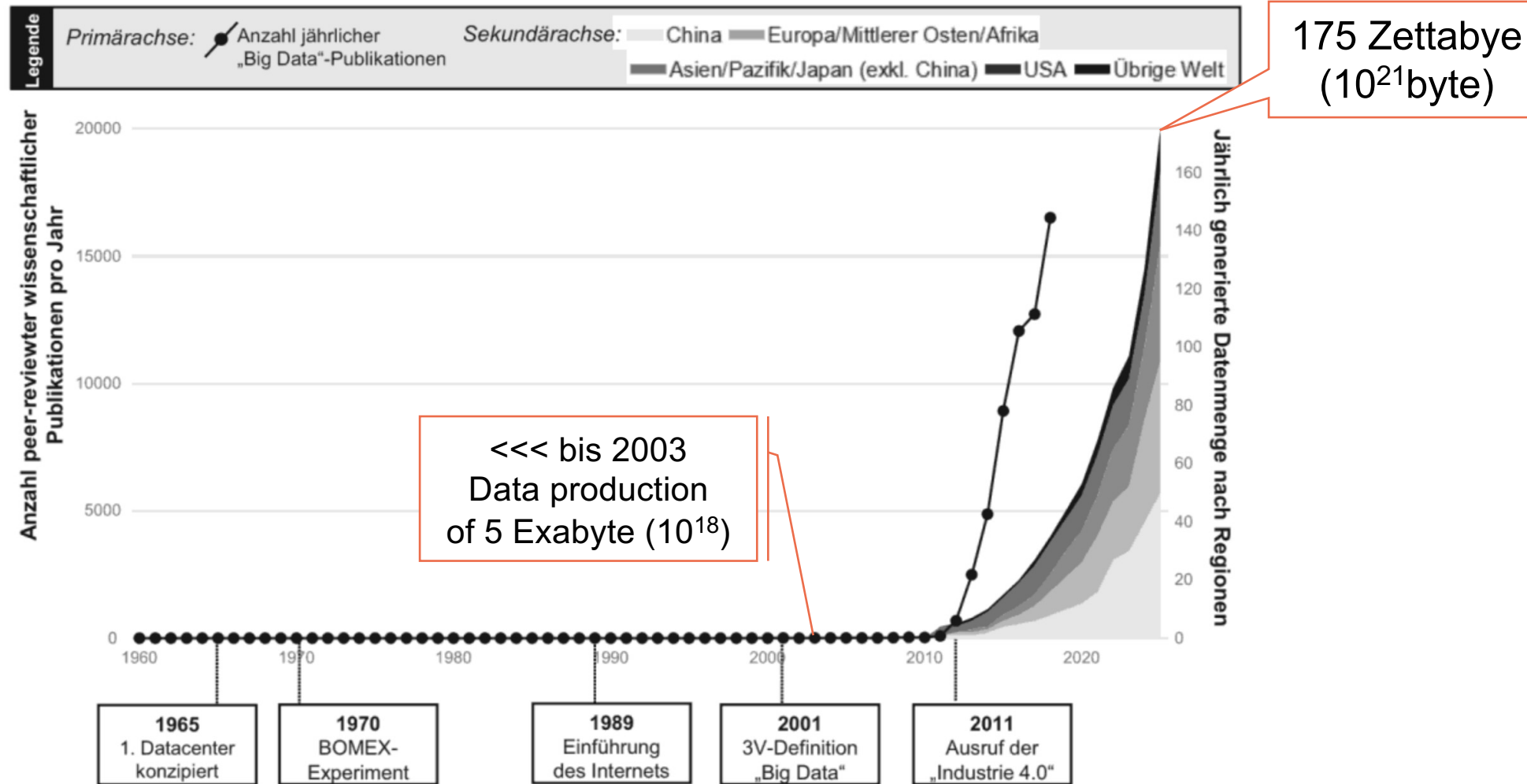
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Source: <http://reports.weforum.org/digital-transformation/wp-content/blogs.dir/94/mp/files/pages/files/dti-executive-summary-20180510.pdf>

Development of the data volumes

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[StKI20:10]



Exercise

To get an impression

- You and an acquaintance have won the lottery.
The prerequisite is that you spend 50,000 euros a day.
- How long will the money last?
 - You win 1 Million
 - The acquaintance wins 1 Billion?

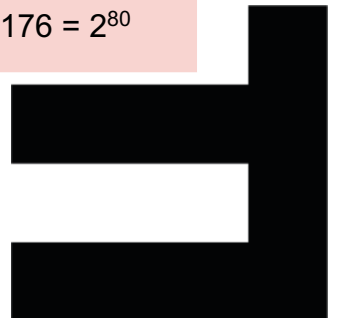


Excursus - data volumes

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Dezimal prefix				Binäre prefixe according to IEC		
Name	Symbol	# of Bytes		Name	Symbol	Anzahl Bytes
Kilobyte	kB	1 000 = 10^3	2,4 %	Kibibyte	KiB	$1\,024^1 = 2^{10}$
Megabyte	MB	1 000 000 = 10^6	4,9 %	Mebibyte	MiB	$1\,024^2 = 1\,048\,576 = 2^{20}$
Gigabyte	GB	1 000 000 000 = 10^9	7,4 %	Gibibyte	GiB	$1\,024^3 = 1\,073\,741\,824 = 2^{30}$
Terabyte	TB	1 000 000 000 000 = 10^{12}	10,0 %	Tebibyte	TiB	$1\,024^4 = 1\,099\,511\,627\,776 = 2^{40}$
Petabyte	PB	1 000 000 000 000 000 = 10^{15}	12,6 %	Pebibyte	PiB	$1\,024^5 = 1\,125\,899\,906\,842\,624 = 2^{50}$
Exabyte	EB	1 000 000 000 000 000 000 = 10^{18}	15,3 %	Exbibyte	EiB	$1\,024^6 = 1\,152\,921\,504\,606\,846\,976 = 2^{60}$
Zettabyte	ZB	1 000 000 000 000 000 000 000 = 10^{21}	18,1 %	Zebibyte	ZiB	$1\,024^7 = 1\,180\,591\,620\,717\,411\,303\,424 = 2^{70}$
Yottabyte	YB	1 000 000 000 000 000 000 000 000 = 10^{24}	20,9 %	Yobibyte	YiB	$1\,024^8 = 1\,208\,925\,819\,614\,629\,174\,706\,176 = 2^{80}$

<https://de.wikipedia.org/wiki/Byte>



Big Data

When is Data **BIG**?

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Typical definition:

- **Big data** is high-volume, high-velocity and/or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation.

Big Data

When is Data BIG??

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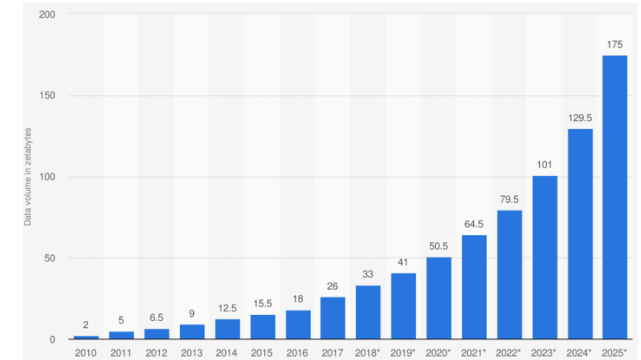
- **Volume:**

The amount of data is extensive and ranges from terabytes to zettabytes.

- **Variety:**

Data used to be mainly from transactional databases. Data types have expanded dramatically to include

- Internet data (e.g. clickstream, social media)
- Research data (e.g. surveys, industry reports)
- Location data (e.g. from Smartphones, geospatial Data)
- Images, videos (e.g. monitoring data, satellite data, diagnostic)
- Metrics (Sensors, RFID-devices)
- Supply-chain Data (EDI, catalogues), E-mail,



Big Data

When is Data BIG??

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- **Velocity:**

The speed at which data is

- created,
- accumulated,
- ingested, and
- processed

- High velocity is attributed to data when we consider the typical speed of transactions on stock exchanges; this speed reaches billions of transactions per day on certain days.
- Today, it is often necessary to react to new data within fractions of a second.
- Processing of streaming data for analysis purposes provide answers to the growing speed of data

[EINa16:945]

Big Data

When is Data BIG??

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- **Value:**

- Data is vital for organisations today
- Data that does not contribute to the organisation's value must be phased out

- **Veracity:**

- Credibility of the source
- Adequacy for the given application

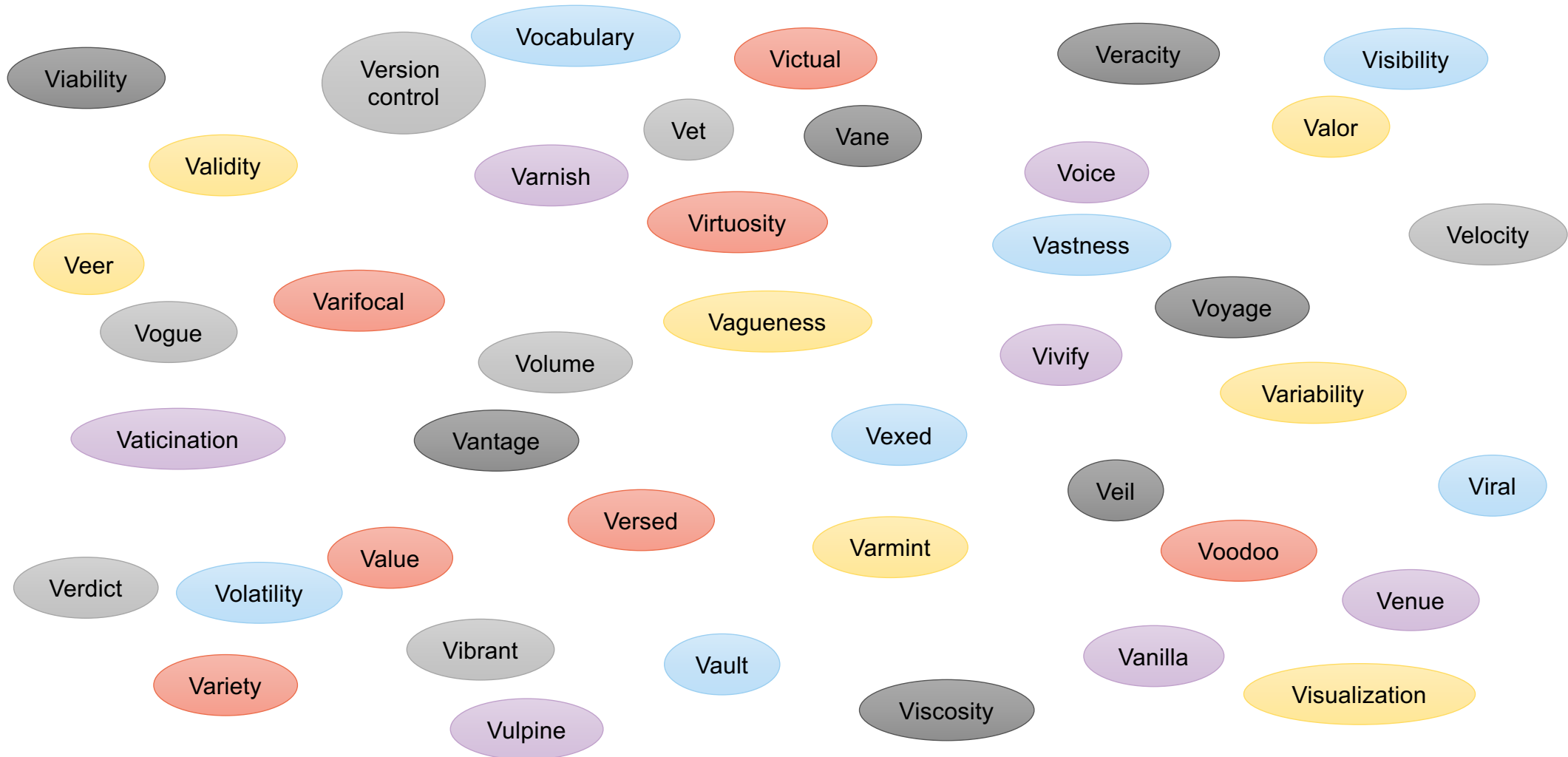
It is closely related to trust. Many sources of data generate data that is uncertain, incomplete, and inaccurate, therefore making its veracity questionable.

- **Variability:**

- Data whose meaning is constantly changing (notably natural language)
- The inconsistencies of the acquired data
- The multitude of data dimensions resulting from multiple disparate data sources
- The inconsistent speed at which the data is provided/loaded

When is Data BIG?? – 42 Vs

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Source: <https://www.elderresearch.com/blog/42-v-of-big-data>

Big Data - Definitionsversuch

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- Today quite common: Definition with 3-6 V (see previous pages)
- Other definition variant:
"[...] big data is any set of data that is too large or too complex to be handled by standard data-processing techniques and typical desktop software."
- Both variants are problematic because they become easily contradictory
- Definition of Big Data as **Complex Data**:
"Big Data ist [...] ein Datenbestand, der eine vergleichsweise hohe Komplexität aufweist. Die Komplexität ergibt sich aus den Anforderungen, die mit der Erfassung, Speicherung oder Verarbeitung der Daten verbunden sind."

Complex data – factors to determine data complexity

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1. **Structure** – Data from different sources, or even different tables from within the same source, could often refer to the same information but be structured entirely differently
2. **Size** – The amount of raw data (tera or petabyte?), **Tall Data** → number of records, or **Wide Data** → number of attributes
3. **Detail** – The level of granularity in which data is explored
If data is delivered as an aggregate, "drill downs" are impossible
4. **Query language** – Different data sources speak different languages (e.g. Pig latin or HiveQL for Hadoop)
5. **Data type** – Structured vs. Semi-/Unstructured, relational vs non-relational
6. **Dispersed data** – Data from different sources (internal/external)
7. **Growth/change rate** – Instability of the structure

Big/Complex Data Challenges

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- Daten storage?
Today especially a cost optimisation question
- Big Data Management →
Ensuring reliable data
- Big Data Acquisition →
Cleaning, transformation, aggregation, encoding, storage and access
- Big Data Aggregation →
Linking of internal and external data
- Imbalanced systems capacities →
Storage capacity vs. speed of I/O operations
- Imbalanced Data →
Unbalanced range of data with different classifications

Big Data – Hadoop



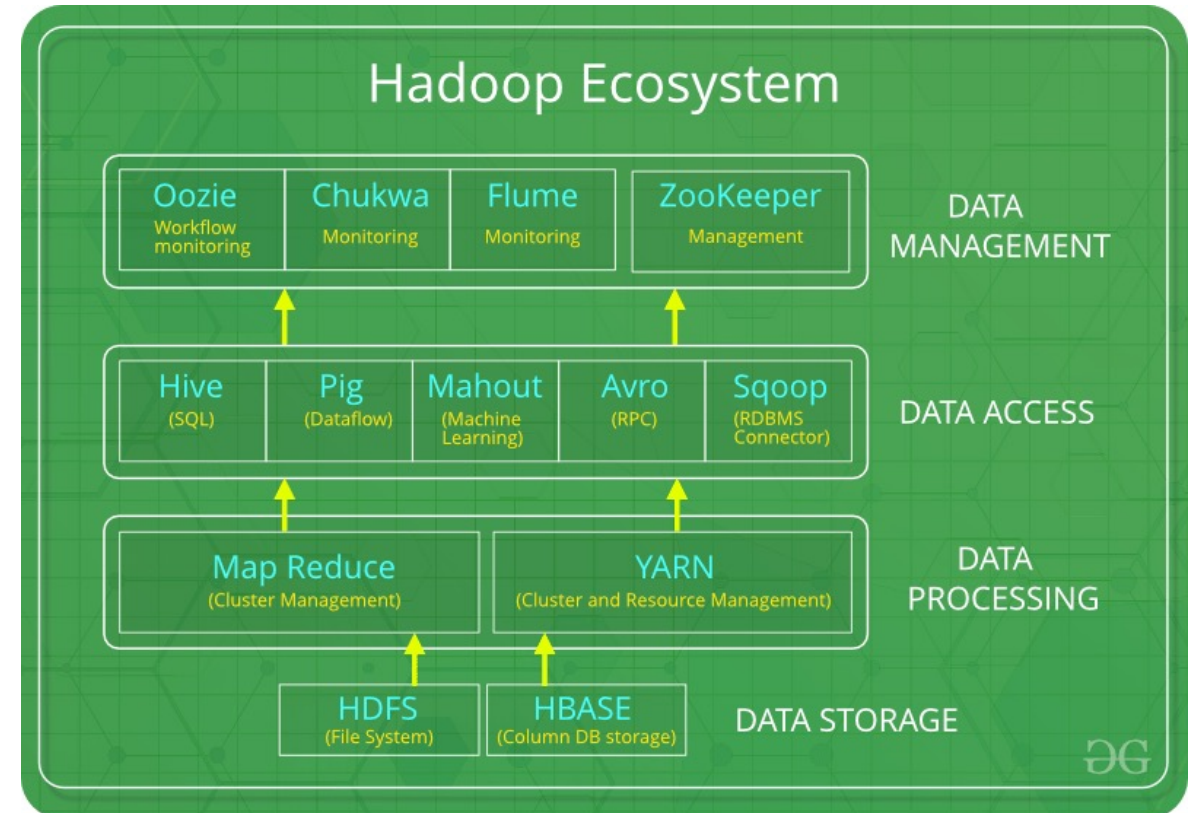
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- Apache Hadoop is an open source a software framework for storage and large scale processing of data-sets on clusters of **commodity hardware**
- The basic principle consists of breaking down large tasks into small subtasks and merging them as aggregates.
 - **Open source:** Apache licence
 - **Highly Scalable:** By adding new nodes and removing depending on demand
 - **Usage of commodity hardware**
 - **Reliable:** Replication of data and compute among multiple nodes

Big Data – Hadoop

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- Key services (examples)
 - **Zookeeper** → Process coordination
 - **Mahout** → Machine Learning Library
 - Query based processing of data services:
Pig → MapReduce, **Hive** → SQL
 - **MapReduce** → high-performance parallel data processing
 - **YARN** – Yet Another Resource Negotiator → Resource management
 - **HDFS** – Hadoop distributed file system → Data storage
 - **Hbase** – non-relational DBMS that uses HDFS as storage



Big Data – HDFS → Hadoop distributed file system

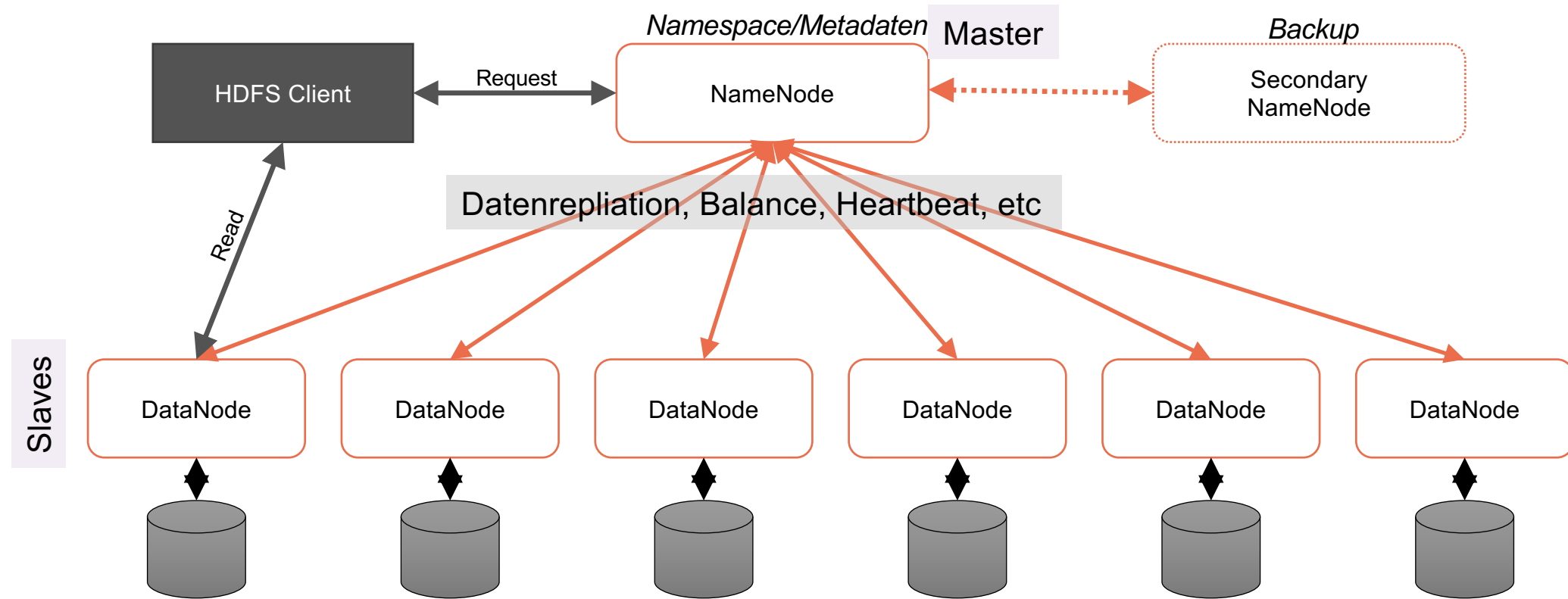
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▪ Assumptions and objectives

- ❑ **Hardware failure:** Using commodity hardware, failure of hardware is the norm rather than an exception. Therefore, with thousands of nodes, automatic detection and recovery from failures becomes a must.
- ❑ **Batch processing:** HDFS has been primarily designed for batch rather than interactive use. High throughput is emphasized over low latency of data access. Full scans of files are typical.
- ❑ **Large datasets:** HDFS was designed to support huge files in the hundreds of gigabytes to terabytes range.
- ❑ **Simple coherency model:** HDFS applications need a one writer and many reader access models for files. File content cannot be updated, but only appended. This model alleviates coherency issues among copies of data.

Big Data – HDFS → Hadoop distributed file system

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Big Data and NoSQL-Systems

- Big Data
- NoSQL Technologies

Why are relational databases not always optimal?

- The relational model is very successful and established worldwide. However, it also has weaknesses that are becoming more and more significant as global data volumes grow.

- **Segmentation of data**

Relational databases typically distribute the data across many tables

(→ normalization, discussed in the earlier lectures)

- For data evaluation, the tables must be merged. These join processes are very resource and time consuming
- The development of the needed SQL statements can be extremely complex
- The design and creation of user-friendly interfaces also tends to be costly on this complex basis



Why are relational databases not always optimal?

- Relational modeling versus object orientation

Object-oriented programming is standard today.

This leads to problems with the relational model – the two ways of thinking/working are fundamentally incompatible

→ Object-relational impedance mismatch

Correctives:

- Object-oriented databases – have not prevailed
- Serialization – conversion of data into strings (e.g. XML)
Cancels all benefits of relational databases
- Object-relational mappers – today's common solution: ready-made libraries that allow good access from programming to the databases.



Why are relational databases not always optimal?

■ Scaling

Scaling affects

- Number of records
- Size of data
- Number of users
- Number of consecutive users
- Complexity of processing instructions

Remark: Conventional tasks rarely reach the limits of current tasks.

Solutions

- Distributed Systems
- noSQL Technologies



- Big data – led by Google, Amazon, Yahoo and Facebook – has spawned some new technologies that go beyond classic RDBMS. They are summarized under the term **NoSQL** – (often interpreted as **Not Only SQL**)
- Some approaches of NoSQL Systems are
 - columnar databases
 - document stores
 - Key-Value/tuple stores
 - graph databases
 - multimodal databases
 - grid and cloud database solutions
 - XML databases
 - multidimensional databases
 - multivalued databases

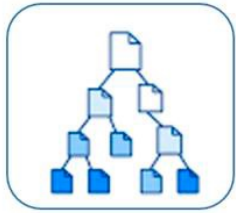


- The term refers to database technologies that do not (or not only) use SQL and do not have a relational data model.
- Basic Demands
 - Simple – non relational – data models
 - Easy data access
 - Excellent horizontal scalability
- NoSQL systems store semi- or unstructured data.
Depending on the given technology, they are optimized for documents, interlinked structures or maximum flexibility.

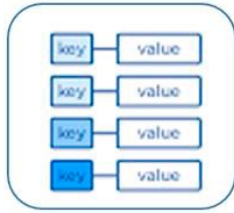


NoSQL – the four most significant models

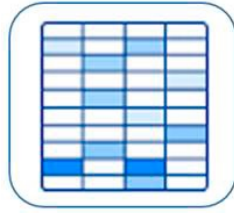
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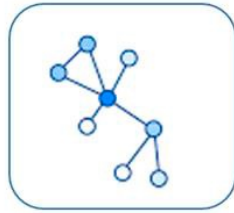
Document Store



Key-Value Store



Wide-Column Store



Graph Store

Model	Characteristics
Key-Value Store	The simplest of the NoSQL databases, data is represented as a collection of key-value pairs
Document Store	Data and metadata are stored hierarchically in JSON-based documents inside the database
Wide-Column Store	Related data is stored as a set of nested-key/value pairs within a single column
Graph Store	Data is stored in a graph structure as node, edge, and data properties

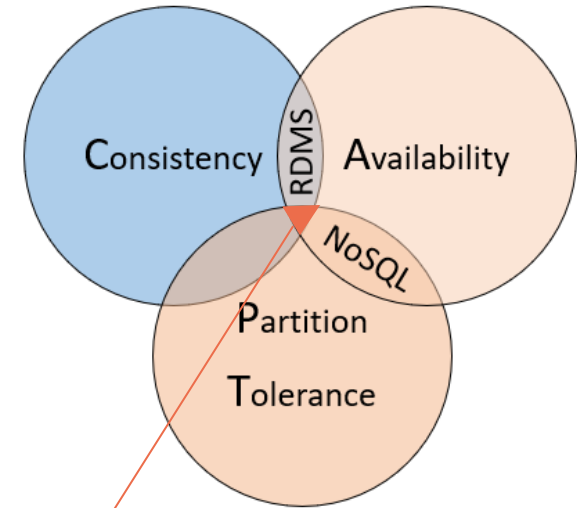
[VeSm22:95]



CAP Theorem

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- Central properties of distributed database systems:
 - **C**onsistency: see ACID
 - **A**vailability: The application is always available and provides timely answers
 - **P**artition Tolerance: If a network node fails or a new node is added, this has no effect on the overall system.
- However, the CAP theorem states that only two of the three properties can be fully satisfied
 - Relational DBMS Meet Consistency and Availability
 - NoSQL systems meet availability and failure tolerance (BASE – Basically Available, Soft State, Eventually Consistent)



Does not exist

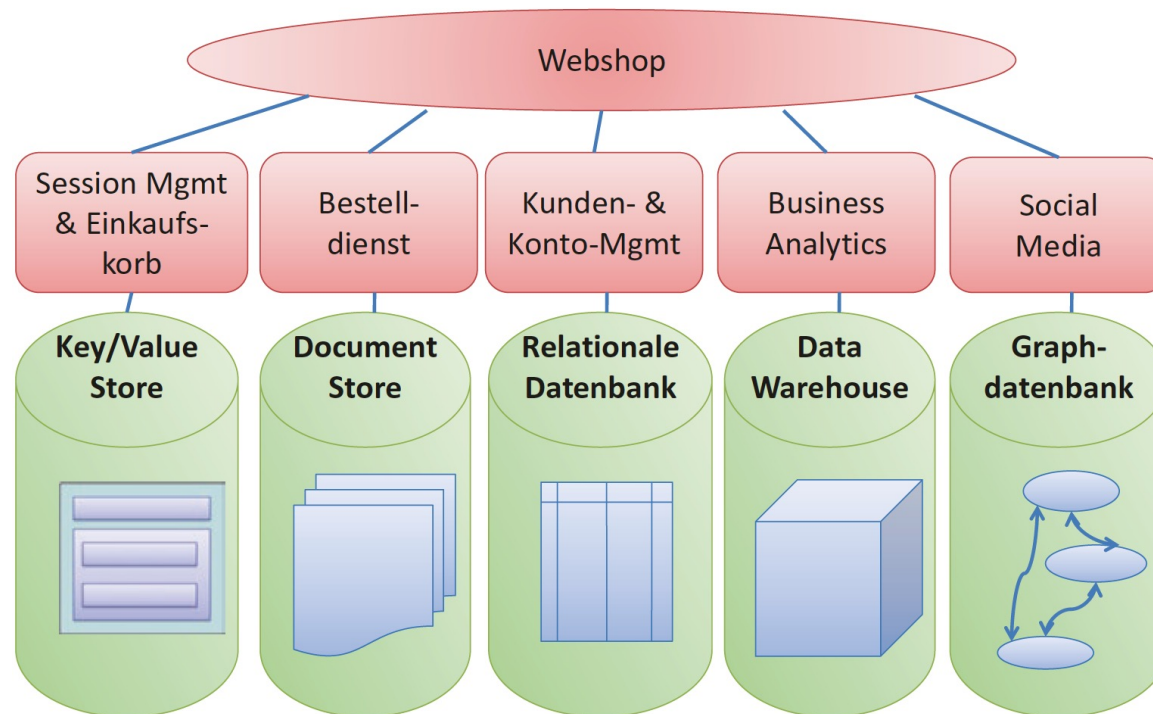
[MeKa19:135] also read <https://www.infoq.com/articles/cap-twelve-years-later-how-the-rules-have-changed/>



NoSQL – Technologies

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- The diversity of approaches means that companies often have to cope with a mix of technologies, but on the other hand can choose the optimal technology for different applications.



[DoMe21:7]

NoSQL – Key-Value store

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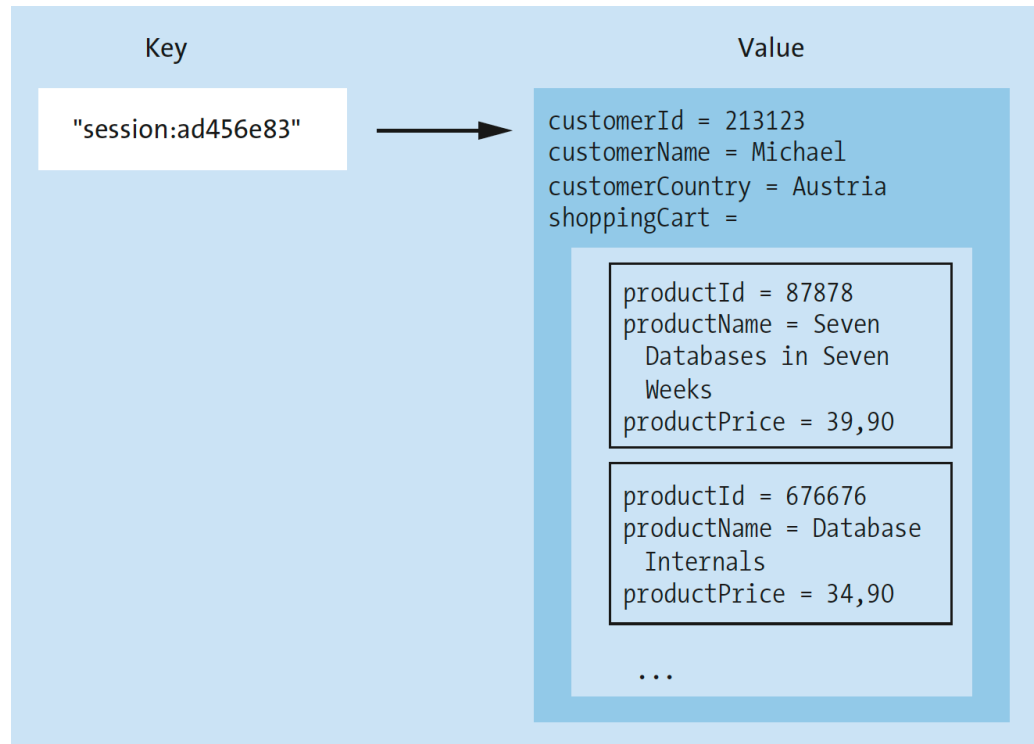
- A key-value storage system (or key-value store) is a system that provides a way to store or update a record (value) with an associated key and to retrieve the record with a given key.
- The structure is very simple, allows easy storage and retrieval of data even of many value pairs, but less complex operations

Key	Value
K1	AAA,BBB,CCC
K2	AAA,BBB
K3	AAA,DDD
K4	AAA,2,01/01/2015
K5	3,ZZZ,5623

NoSQL – Key-Value store

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- The values can also become quite complex. The evaluation does not take place in the store, since – in its pure form – key-value stores do not "know" the content of the data.
- There are key-value storage systems require the stored data to follow a specified data representation, allowing the data store to interpret the stored values and execute simple queries based on stored values.



NoSQL – Document store

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- Document stores are extensions of key-value stores
- They contain a unique key and values with complex data structures (the border to the key-value stores is therefore fluid)
- Unlike relational databases, the structures do not have to be implemented in advance, but can grow dynamically. Indexing and filter functions allow better access to content as with conventional key-value stores

NoSQL – Document store

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■ Example: Populating an Address Database with JSON

```
■ db.adressbuch.insert(  
  {  
    vorname : „Heidi“,  
    nachname : „Weber“,  
    email : „hwe@fhv.at“,  
    adresse : {  
      strasse : „Hochschulstraße 1“,  
      ort : „Dornbirn“,  
      plz : 6850,  
      land : „Austria“  
    },  
    korrespondenzsprache : „deutsch“  
  }  
)
```

```
db.adressbuch.insert(  
  {  
    vorname : „Max“,  
    nachname : „Mustermann“,  
    mobile : „001798975644“  
  }  
)
```

```
db.adressbuch.insert(  
  {  
    vorname : „Sabine“,  
    nachname : „Schmidt“,  
    linkedin : „linkedin.com/sschmidt“  
  }  
)
```

Important:
Each of these records has a different structure, but they can be stored and searched together in the store.

NoSQL – Wide-Column Stores

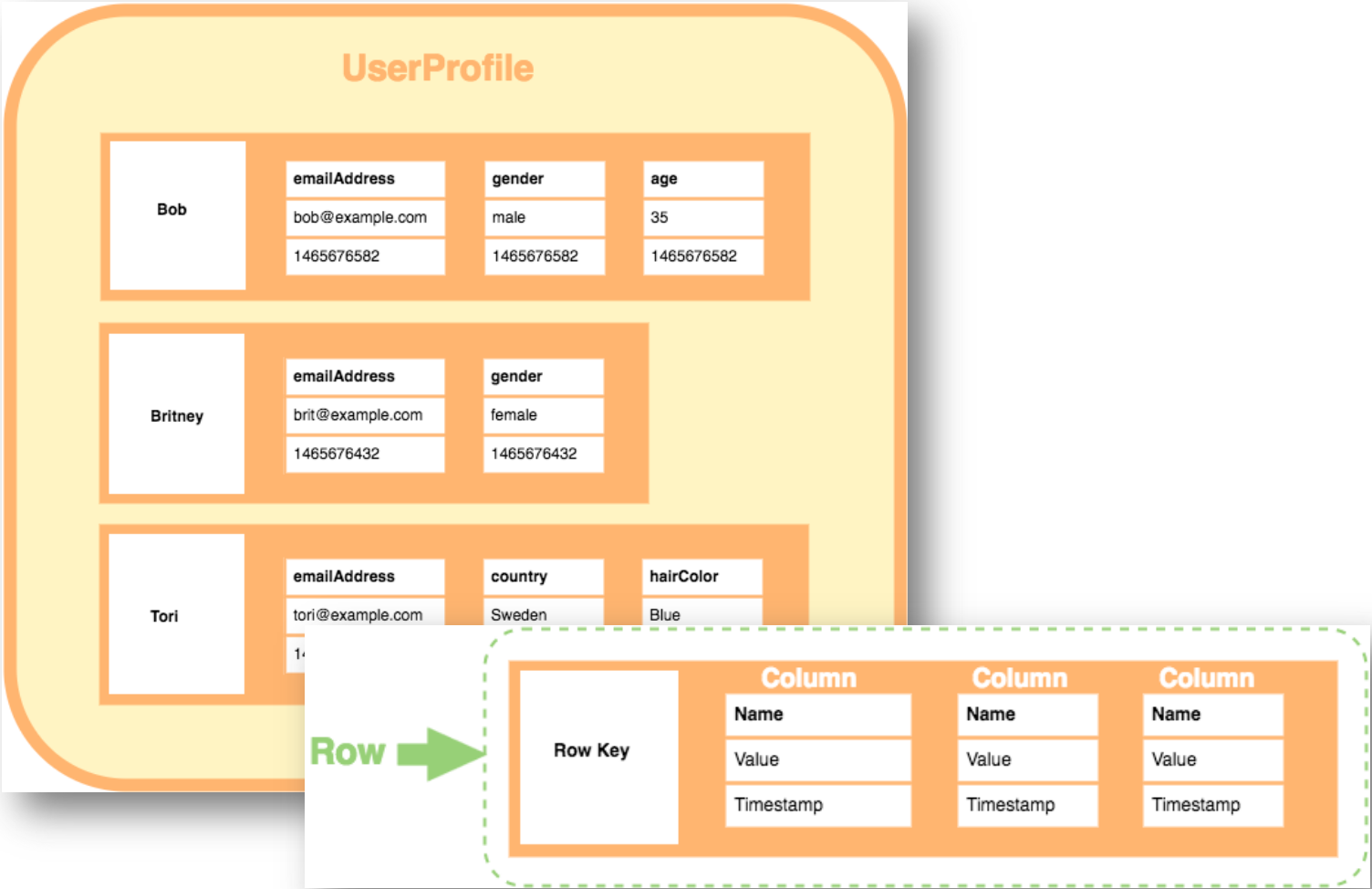
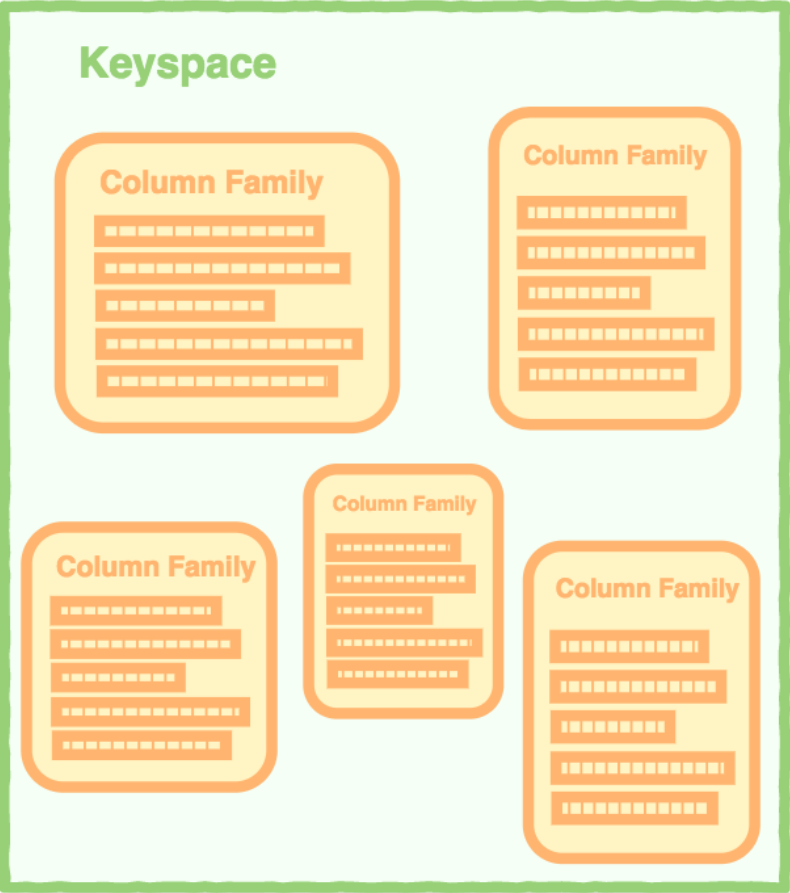
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- Wide column or column family systems store records in multidimensional maps that are similar to relational objects.
- Related data is grouped in a column family that contains any number of rows.
- Each row has a row key and any number (1 to millions) entries of key-value pairs.
- As a third dimension, time stamps and corresponding value variations can also be created.
- The storage structure thus corresponds to that of a key-value store, but with (mostly) multidimensional key
- Wide column stores are particularly suitable for
 - Complex and typically schema-free object structures
 - Fast read and write operations of atomic data sets
 - example: Storage of event series

[FaMe16:118,122]

NoSQL – Wide-Column Systeme

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[<https://database.guide/what-is-a-column-store-database/>]

NoSQL – Columnar systems

- Columnar or column-oriented systems store their data in columns, unlike relational systems, which store rows (records) sequentially.
- Each attribute of a relation is stored separately, with values of the attribute from successive tuples stored at successive positions in the file.
- Data compression of columnar systems is very effective because the data-types do not change
- Column-oriented storage is well suited for data analysis queries, which process many rows of a relation, but often only access some of the attributes.

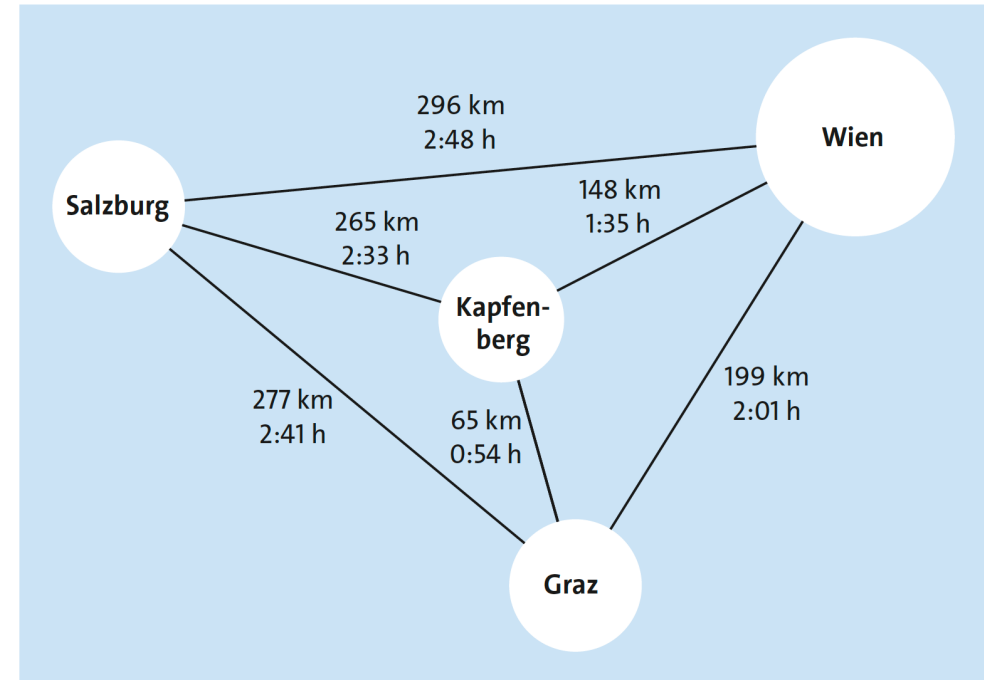
Column-oriented and wide-column systems are easily confused, but are structurally completely different

10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
83821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

Graph-based databases

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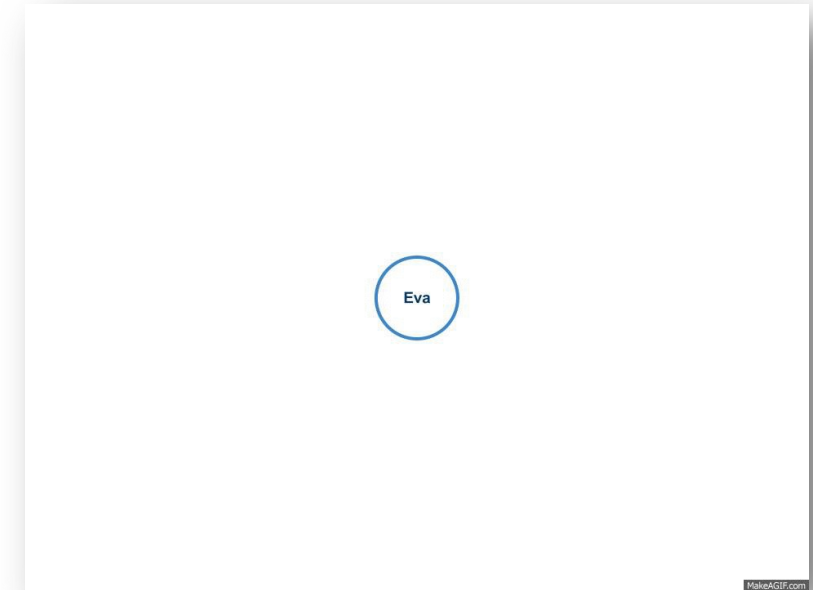
- Graphs are linked data structures composed of nodes and edges and their respective properties.
 - **Nodes** describe the entities that are connected to each other. These can be, for example, places or people.
 - **Edges** establish the connection between the objects. They can be directed (e.g. pedigree) or non-directed.
 - **Properties/Weight:**
Nodes and edges can be weighted and have properties. This allows you to express e.g. the number of inhabitants of a city, the contact details of a person or the costs associated with a geographical distance (time, distance).



Graph-based databases

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- Graphs are a good way of representing relationships between entities.
- For example, social graphs represent social networks and the relationships of the people involved (e.g. people, photos, places and interests).
- In graph-based systems, each node "knows" who its neighbours are and how it is connected to them (edge).
- Additional nodes do not create a relevant increase in query complexity.
- Typical tasks, such as the shortest distance between two accounts, can be calculated efficiently. The query always starts at a node and moves through the network from there.



[FaMe16:122] <https://www.facebook.com/notes/822349768589276/>

Applications

■ Key-Value Stores

- Volatile temporary storage for websites and other applications.
- Persistent storage of records with very simple data structures, which can be represented as a key-value pair.

■ Document Stores

- Simple databases with objects with different schemas (for example: HTML pages, XML document, address books).
- Databases that only create simple relations between documents, i.e. no complex product creation between the documents.

■ Column Family Stores

- Complex and preferably schema-free object structures
- Fast read and write operations on atomic data sets

■ Spaltenorientierte Systeme

- Business Intelligence und Analytics on large amounts of data

■ Graphen Datenbanken

- Graph problems such as people relationships, network topologies, timetables and check them.



SQL or NoSQL?

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Consider a NoSQL datastore when:	Consider a relational database when:
You have high volume workloads that require predictable latency at large scale (e.g. latency measured in milliseconds while performing millions of transactions per second)	Your workload volume generally fits within thousands of transactions per second
Your data is dynamic and frequently changes	Your data is highly structured and requires referential integrity
Relationships can be de-normalized data models	Relationships are expressed through table joins on normalized data models
Data retrieval is simple and expressed without table joins	You work with complex queries and reports
Data is typically replicated across geographies and requires finer control over consistency, availability, and performance	Data is typically centralized, or can be replicated regions asynchronously
Your application will be deployed to commodity hardware, such as with public clouds	Your application will be deployed to large, high-end hardware

[VeSm22:98]

...or NewSQL

- According to the CAP theorem, NoSQL databases do not fulfil the ACID properties that are essential for transactional databases.
- Technologies under the term NewSQL are trying to find answers to this:
- Two variants are prevalent
 - Relational database systems that respond better to the classic weaknesses (see above) of RDMBS, especially horizontal scaling, but still strictly adhere to the ACID rules for transactions.
 - Non-relational systems (NoSQL) that allow the use of SQL and strive for the highest possible consistency (better subsumed under Not Only SQL).

📖 First mention of NewSQL in 2011
<https://cs.brown.edu/course/cs227/archives/2012/papers/newsql/aslett-newsql.pdf>



Additional Literature and Lecture Sources

27 November 2023

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